Roadmap to Become an Agentic AI Developer

Phase 1: Strengthen Programming Fundamentals (1–2 Months)
- Python: Functions, OOP, File Handling, Exception Handling, Modules, Virtual Environments
- SQL: Joins, Subqueries, Views, Stored Procedures, Indexes
- Git & GitHub: Repositories, Branching, Pull Requests, Version Control

Phase 2: Learn AI & Machine Learning Basics (1–2 Months)
- AI, Machine Learning, Deep Learning
- Supervised and Unsupervised Learning
- Neural Networks
- Libraries: NumPy, Pandas, Matplotlib, Scikit-learn
- Projects: Student Marks Predictor, House Price Predictor, Spam Classifier

Phase 3: Learn NLP & LLM Fundamentals (1 Month)
- Tokenization, Embeddings, Transformers, Attention Mechanism
- Prompt Engineering
- Hugging Face and OpenAI APIs

Phase 4: Build LLM Applications (1–2 Months)
- PDF Question Answering Bot
- Resume Analyzer
- Document Summarizer
- Customer Support Chatbot

Phase 5: Learn RAG (Retrieval-Augmented Generation)
- Embeddings
- Vector Databases
- Semantic Search
- Tools: ChromaDB, Pinecone

Phase 6: Learn Agent Frameworks
- LangChain
- CrewAI
- Microsoft AutoGen

Phase 7: Learn Tool Calling & Automation
- Weather API
- Email API
- Calendar API
- Database Queries

- Projects: Travel Planner, Appointment Scheduler, Email Assistant

Phase 8: Build Real Agentic AI Projects
- Study Assistant
- To-do Manager
- Research Agent
- Resume Screening Agent
- Customer Support Agent
- AI Coding Assistant

Phase 9: Deployment
- FastAPI
- Docker
- Linux Basics
- Cloud Platforms: AWS, Google Cloud, Azure

Phase 10: Build a Portfolio
- PDF Chatbot
- RAG Knowledge Assistant
- AI Research Agent
- Multi-Agent Workflow System
- AI Email Assistant
- Django + AI Integration Project

Suggested Timeline:
Month 1–2: Advanced Python, Git/GitHub, APIs
Month 3: Machine Learning and NLP Basics
Month 4: OpenAI APIs, Prompt Engineering, Chatbot Projects
Month 5: RAG and Vector Databases
Month 6: LangChain or CrewAI, First AI Agent
Month 7–8: Multi-Agent Systems, Deployment, Portfolio Projects